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Publications by Jared Miller

I. Journal Papers (published)

1) J. Miller, F. Jakob, C. Scherer, and A. Iannelli, "Analysis and synthesis of switched optimization algorithms," *Nonlinear Analysis: Hybrid Systems*, vol. 62, p. 101788, 2026. [link]
2) J. Miller and M. Sznaier, "Quantifying the safety of trajectories by the data corruption needed to crash," *SIAM Journal on Control and Optimization*, vol. 64, no. 4, pp. 2204–2223, 2026. [link]
3) J. Miller, M. Tacchi, D. Henrion, and M. Sznaier, "Unsafe probabilities and risk contours for stochastic processes using convex optimization," *Automatica*, vol. 188, p. 112939, 2026. [link]
4) J. Miller, T. Dai, M. Sznaier, and B. Shafai, "Data-Driven Control of Switched Linear Positive Systems using Robust Convex Optimization," *IEEE Transactions on Automatic Control*, pp. 1–8, 2026. [link], Early Access
5) J. Miller and M. Sznaier, "Peak estimation of hybrid systems with convex optimization," *Automatica*, vol. 183, p. 112641, 2026. [link]
6) J. Miller, J. Wang, and F. Guo, "Sparse polynomial matrix optimization," *SIAM Journal on Optimization*, vol. 36, no. 2, pp. 503–533, 2026. [link]
7) T. Imtiaz, M. R. Kohler, J. F. Miller, Z. Wang, M. Eskandar, M. Sznaier, O. Camps, and J. Dy, "SAIF: Sparse adversarial and imperceptible attack framework," *Transactions on Machine Learning Research*, 2025. [link]
8) J. Miller and M. Sznaier, "Analysis of Input-Affine Dynamical Systems using Parameterized Robust Counterparts," *IEEE Transactions on Automatic Control*, vol. 70, no. 8, pp. 5413–5428, 2025. [link]
9) J. Miller, M. Tacchi, M. Sznaier, and A. Jasour, "Convex Computation of Value-at-Risk Bounds for Stochastic Processes," *IEEE Transactions on Automatic Control*, vol. 70, no. 6, pp. 3936–3951, 2025. [link]
10) J. Miller, T. Dai, and M. Sznaier, "Robust Data-Driven Control of Discrete-Time Linear Systems with Errors in Variables," *IEEE Transactions on Automatic Control*, vol. 70, no. 2, pp. 947–962, 2025. [link]
11) J. Miller and R. S. Smith, "Peak estimation of rational systems using convex optimization," *European Journal of Control*, p. 101088, 2024. [link] (2024 European Control Conference Special Issue)
12) J. Miller, T. Dai, and M. Sznaier, "Data-Driven Superstabilizing Control Under Quadratically-Bounded Errors-in-Variables Noise," *IEEE Control Systems Letters*, vol. 8, pp. 1655–1660, 2024. [link]
13) J. Zheng, J. Miller, and M. Sznaier, "Data-Driven Safe Control of Discrete-Time Non-Linear Systems," *IEEE Control Systems Letters*, vol. 8, pp. 1553–1558, 2024. [link]
14) J. Miller and M. Sznaier, "Bounding the Distance to Unsafe Sets with Convex Optimization," *IEEE Transactions on Automatic Control*, vol. 68, no. 12, pp. 7575 – 7590, 2023. [link]
15) J. Zheng, T. Dai, J. Miller, and M. Sznaier, "Robust Data-Driven Safe Control using Density Functions," *IEEE Control Systems Letters*, vol. 7, pp. 2611–2616, 2023. [link] (LCSS-CDC)
16) J. Miller and M. Sznaier, "Data-Driven Gain Scheduling Control of Linear Parameter-Varying Systems using Quadratic Matrix Inequalities," *IEEE Control Systems Letters*, vol. 7, pp. 835–840, 2022. [link] (LCSS-ACC)
17) J. Miller, Y. Zheng, M. Sznaier, and A. Papachristodoulou, "Decomposed structured subsets for semidefinite and sum-of-squares optimization," *Automatica*, vol. 137, pp. 110–125, 2022. [link]
18) J. Miller, D. Henrion, and M. Sznaier, "Peak Estimation Recovery and Safety Analysis," *IEEE Control Systems Letters*, vol. 5, no. 6, pp. 1982–1987, 2021. [link] (LCSS-ACC)
19) J. Miller, M. A. Al-Radhawi, and E. D. Sontag, "Mediating Ribosomal Competition by Splitting Pools," *IEEE Control Systems Letters*, vol. 5, no. 5, pp. 1555–1560, 2021. [link] (LCSS-ACC)

II. Journal Papers (accepted)

1) J. Miller, M. Bianchi, and F. Dörfler, "Convergence and Robustness Bounds for Distributed Asynchronous Shortest-Path," 2026. [link], (Transactions on Control of Network Systems)
2) J. Miller, "Incremental Peak-to-Peak Gain Computation and Regulation of Discrete-Time Positive Luré Systems using Linear Programming ," 2026. [link], (Automatica, Technical Note)

III. Journal Papers (submitted)

1) J. Miller, C. Scherer, F. Jakob, and A. Iannelli, "Structure, Analysis, and Synthesis of First-Order Algorithms," 2026. [link]
2) N. Schmid, J. Miller, T. Zeller, M. Fochesato, T. Sutter, and J. Lygeros, "Joint Chance Constrained Safe-Optimal Control," 2026. [link]
3) V. Behrunani, C. Koepele, J. Miller, A. Aboudonia, P. Heer, R. S. Smith, and J. Lygeros, "Plug and play distributed control of clustered energy hub networks," 2025. [link]
4) C. Meroni, J. Miller, and M. Velasco, "Approximation of starshaped sets using polynomials," 2025. [link]
5) J. Miller, C. Meroni, M. Tacchi, and M. Velasco, "Maximizing Slice-Volumes of Semialgebraic Sets using Sum-of-Squares Programming," 2024. [link]
6) J. Miller, N. Schmid, M. Tacchi, D. Henrion, and R. S. Smith, "Peak Time-Windowed Risk Estimation of Stochastic Processes," 2024. [link]
7) D. Henrion, J. Miller, and M. S. E. Din, "Algebraic Proofs of Path Disconnectedness using Time-Dependent Barrier Functions," 2024. [link]

IV. PhD Thesis

1) J. Miller, *Safety Quantification for Nonlinear and Time-Delay Systems using Occupation Measures*. PhD thesis, Northeastern University, April 2023. [link], [interview for Ph.D.s in Control]

V. Conference Proceedings (published)

1) J. Miller and P. Karamanakos, "Optimal Pulse Patterns through a Hybrid Optimal Control Perspective," 2025. [link] (2025 IEEE Conference on Decision and Control)
2) J. Miller, J. Eising, F. Dörfler, and R. S. Smith, "Data-driven structured robust control of linear systems," in *2025 European Control Conference (ECC)*, pp. 96–101, 2025. [link]
3) P. F. De Paola, J. Miller, A. Borri, A. Paglialonga, and F. Dabbene, "Robust control-driven counterfactual generation for uncertain systems," *11th IFAC Symposium on Robust Control Design ROCOND 2025*, vol. 59, no. 16, pp. 116–121, 2025. [link]
4) J. Miller, N. Schmid, M. Tacchi, D. Henrion, and R. S. Smith, "Peak time-windowed mean estimation using convex optimization," in *2024 IEEE 63rd Conference on Decision and Control (CDC)*, pp. 703–708, 2024. [link]
5) J. Miller, J. Zheng, M. Sznaier, and C. Hixenbaugh, "Data-driven superstabilization of linear systems under quantization," in *2024 American Control Conference (ACC)*, pp. 1146–1151, 2024. [link]

6) A. Honarpisheh, R. Singh, J. Miller, and M. Sznaier, "Identification of low order systems in a loewner framework," in *20th IFAC Symposium on System Identification SYSID 2024*, vol. 58, pp. 199–204, 2024. [link]
7) M. Abdalmoaty, J. Miller, M. Yin, and R. S. Smith, "Frequency-domain identification of discrete-time systems using sum-of-rational optimization," in *20th IFAC Symposium on System Identification SYSID 2024*, vol. 58, pp. 121–126, 2024. [link]
8) J. Miller, M. Korda, V. Magron, and M. Sznaier, "Peak Estimation of Time Delay Systems Using Occupation Measures," in *2023 62nd IEEE Conference on Decision and Control (CDC)*, pp. 5294–5300, 2023. [link]
9) J. Miller, T. Dai, M. Sznaier, and B. Shafai, "Data-Driven Control of Positive Linear Systems using Linear Programming," in *2023 62nd IEEE Conference on Decision and Control (CDC)*, pp. 1588–1594, 2023. [link]
10) J. Miller, M. Tacchi, M. Sznaier, and A. Jasour, "Peak Value-at-Risk Estimation for Stochastic Differential Equations Using Occupation Measures," in *2023 62nd IEEE Conference on Decision and Control (CDC)*, pp. 4836–4842, 2023. [link]
11) J. Miller, T. Dai, and M. Sznaier, "Superstabilizing Control of Discrete-Time ARX Models under Error in Variables," in *22nd IFAC World Congress*, vol. 56, pp. 2444–2449, 2023. [link]
12) J. Miller, T. Dai, and M. Sznaier, "Data-Driven Superstabilizing Control of Error-in-Variables Discrete-Time Linear Systems," in *2022 61st IEEE Conference on Decision and Control (CDC)*, pp. 4924–4929, 2022. [link]
13) J. Miller and M. Sznaier, "Bounding the Distance of Closest Approach to Unsafe Sets with Occupation Measures," in *2022 61st IEEE Conference on Decision and Control (CDC)*, pp. 5008–5013, 2022. [link]
14) F. Bečanović, J. Miller, V. Bonnet, K. Jovanović, and S. Mohammed, "Assessing the Quality of a Set of Basis Functions for Inverse Optimal Control via Projection onto Global Minimizers," in *2022 IEEE 61st Conference on Decision and Control (CDC)*, pp. 7598–7605, 2022. [link]
15) J. Miller and M. Sznaier, "Facial Input Decompositions for Robust Peak Estimation under Polyhedral Uncertainty," *IFAC-PapersOnLine*, vol. 55, no. 25, pp. 55–60, 2022. [link]
16) J. Miller, D. Henrion, M. Sznaier, and M. Korda, "Peak Estimation for Uncertain and Switched Systems," in *2021 60th IEEE Conference on Decision and Control (CDC)*, pp. 3222–3228, 2021. [link]
17) J. Miller, R. Singh, and M. Sznaier, "MIMO System Identification by Randomized Active-Set Methods," in *2020 59th IEEE Conference on Decision and Control (CDC)*, pp. 2246–2251, 2020. [link]
18) J. Miller, Y. Zheng, M. Sznaier, and A. Papachristodoulou, "Decomposed Structured Subsets for Semidefinite Optimization," in *2020 21st IFAC World Congress*, 2020. [link]
19) C. Wu, J. Miller, Y. Chang, M. Sznaier, and J. Dy, "Solving Interpretable Kernel Dimensionality Reduction," in *Advances in Neural Information Processing Systems* (H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. Fox, and R. Garnett, eds.), vol. 32, pp. 7915–7925, Curran Associates, Inc., 2019. [link]
20) J. Miller, Y. Zheng, B. Roig-Solvas, M. Sznaier, and A. Papachristodoulou, "Chordal Decomposition in Rank Minimized Semidefinite Programs with Applications to Subspace Clustering," in *2019 IEEE 58th Conference on Decision and Control (CDC)*, pp. 4916–4921, 2019. [link]
21) J. Miller and B. Shafai, "A Model of Heave Dynamics for Bagged Air Cushioned Vehicles," in *2019 IEEE Conference on Control Technology and Applications (CCTA)*, pp. 976–981, 2019. [link]
22) B. Taskazan, J. Miller, U. Inyang-Udoh, O. Camps, and M. Sznaier, "Domain Adaptation Based Fault Detection in Label Imbalanced Cyberphysical Systems," in *2019 IEEE Conference on Control Technology and Applications (CCTA)*, pp. 142–147, 2019. [link]

VI. Conference Proceedings (accepted)

1) J. Miller, N. Estay, P. Karamanakos, and A. Angulo, "Symmetry-Reduced Algebraic Selective Harmonic Elimination for Cascaded H-Bridge Converters with Unequal DC Source Levels," in *Proceedings of the Energy Conversion Congress & Expo: Europe*, 2026

VII. Preprints

1) J. Miller, P. Karamanakos, and T. Geyer, "Bounding the Minimal Current Harmonic Distortion in Optimal Modulation of Single-Phase Power Converters," 2025. [link]

VIII. Seminars

1) "Structure, Analysis, and Synthesis of First-Order Algorithms," April 13, 2026, Technical University of Hamburg, Hamburg, DE.
2) "Structure, Analysis, and Synthesis of First-Order Algorithms," March 30, 2026, Lund University, Lund, SE.
3) "Structure, Analysis, and Synthesis of First-Order Algorithms," March 27, 2026, KTH, Stockholm, SE [link].
4) "Structure, Analysis, and Synthesis of First-Order Algorithms," March 18, 2026, 96th GAMM Annual Meeting, University of Stuttgart, Stuttgart, DE.
5) "Structure, Analysis, and Synthesis of First-Order Algorithms," March 12, 2026, University of Lorraine, Nancy, DE.
6) "Structure, Analysis, and Synthesis of First-Order Algorithms," March 3, 2026, Liebniz University Hannover, Hannover, DE.
7) "Structure, Analysis, and Synthesis of First-Order Algorithms," February 20, 2026, INRIA, Paris, FR.
8) "Are optimal pulse patterns optimal? Bounding minimal harmonic distortion of DC/AC power converters," November 28, 2025, Control Systems Talks, FAU Erlangen-Nürnberg, DE [link].
9) "Second-Order Cone Programming in Measures," June 23, 2025. European Conference on Operational Research, Leeds, UK.
10) "Are optimal pulse patterns optimal? Bounding minimal harmonic distortion of DC/AC power converters," May 14, 2025, Control Systems Colloquium, Eindhoven, NL.
11) "Frequency-Domain Identification of Discrete-Time Systems using Sum-of-Rational Optimization," August 19-23, 2024, 26th International Symposium on Mathematical Theory of Networks and Systems, Cambridge, UK. [link]
12) "Analysis and Control of Input-Affine Systems using Parameterized Robust Counterparts," August 19-23, 2024, 26th International Symposium on Mathematical Theory of Networks and Systems, Cambridge, UK. [link]
13) "Going beyond ODE systems with the moment-SOS hierarchy," July 28-August 2, 2024, Polynomial Optimization for Nonlinear Dynamics: Theory, Algorithms, and Applications, Oberwolfach, DE. [link]
14) "Risk Analysis of Stochastic Processes using Occupation Measures," April 11, 2024, H-CODE Seminar Series, Laboratory of Signals and Systems (L2S), Paris-Saclay, FR.
15) "Maximizing Slice-Volumes of Semialgebraic Sets using Sum-of-Squares Programming," March 13, 2024, Moments and Polynomials: Applications and Theory Universitz of Konstanz, Konstanz, DE. [link]
16) "Data-Driven Safety Quantification using Robust Optimization," February 16, 2024, Institute for Systems Theory and Automatic Control (IST), University of Stuttgart, Stuttgart, DE. [link]
17) "Data-Driven Safety Quantification using Robust Optimization," October 18, 2023, Cybernetic Systems and Controls Lab (CSCL), Arizona State University, Tempe, AZ. [link].

18) "Risk analysis for stochastic processes using polynomial optimization," October 15-18, 2023, Convex Relaxations for Polynomial Optimization, INFORMS Annual Meeting, Phoenix, AZ [link].
19) "Data-Driven Safety Quantification using Robust Optimization," October 11, 2023, Safe Autonomy and Intelligent Distributed Systems (SAIDS) group, University of Southern California, Los Angeles, CA. [link]
20) "Data-Driven Safety Quantification using Infinite-Dimensional Robust Convex Optimization," July 27, 2023, Konstanz Real Algebraic Geometry Seminar, University of Konstanz. [link]
21) "Data-Driven Safety Quantification using Infinite-Dimensional Robust Convex Optimization", May 29, 2023, Student Seminar Series on Optimization, Control & Learning, UC San Diego. [link]
22) "Quantifying Safety under Uncertainty using Occupation Measures", May 26, 2023, Control Seminars @ UCI, UC Irvine
23) "Data-Driven Safety Quantification using Infinite-Dimensional Robust Convex Optimization", May 19, 2023, Multi-Robot Systems Lab Meeting, Stanford University. [link]
24) "Analysis and Control of Time-Delay Systems Using Polynomial Optimization", May 14, 2023, MS14 Studying Dynamics using Polynomial Optimization Tools, SIAM Conference on Dynamical Systems. [link]
25) "Data-Driven Control under Input and Measurement Noise", April 9, 2023, Oden Institute Seminar, UT Austin. [link]
26) "Safety Quantification for Nonlinear and Time-Delay Systems using Occupation Measures", April 3, 2023, PhD Thesis Defense, Northeastern University. [link]
27) "Data-Driven Control under Input and Measurement Noise", NYU MERIIT Lab Seminar Series, New York City, Feb 21, 2023. [link]
28) "Bounding the Distance to Unsafe Sets with Convex Optimization", DCSD Rising Stars, 2nd Modeling, Estimation and Control Conference, Jersey City, October 2-5 2022. [link]
29) "Bounding distances to unsafe sets", June 16, 2022, IfA Coffee Talks, ETH Zurich. [link]
30) "Bounding distances to unsafe sets", June 14, 2022, LA3 Meeting, EPFL Lausanne. [link]
31) "Bounding distances to unsafe sets", June 3, 2022, Journées SMAI MODE, University of Limoges (XLIM). [link]
32) Tutorials about Interior Point Methods, Polynomial Optimization, Frank-Wolfe algorithms and variations, and SDP approximations, May 16-20, Sparsity and Big Data in Control, Systems Identification, and Machine Learning, European Embedded Control Institute.
33) "Bounding distances to unsafe sets", April 14, 2022, Conic Linear Optimization for Computer-Assisted Proofs, Mathematisches Forschungsinstitut Oberwolfach (MFO). [link]
34) "Bounding distances to unsafe sets", June 28, 2021, Brainstorming days on measure and polynomial optimization (BrainPOP), LAAS-CNRS. [link]
35) "Data-Driven Peak and Reachability Set Estimation", May 25, 2021, MS112 Methods of Learning Dynamical Systems for Control, SIAM Conference on Dynamical Systems. [link]
36) "Analysis and Control of Time-Delay Systems with Occupation Measures", May 3, 2021, BrainPOP, LAAS-CNRS. [link]
37) "Exploiting Structure in Rank-Constrained and Approximated Semidefinite Programs", December 19, 2019, TISEM Operations Research Seminar, Tilburg University. [link]

IX. Poster Sessions

1) "Are Optimal Pulse Patterns Optimal? Bounding harmonic distortion in power converters." September 2, 2025. Swiss CLOCK summit, Engelberg (Best Poster Award, juried). [link]
2) "Quantifying Trajectory Safety by the Minimum Data Corruption Needed to Crash." September 30-October 2, 2024. Symposium on Systems Theory in Data and Optimization. Universität Stuttgart. [link]
3) "Risk Analysis of Stochastic Processes using Polynomial Optimization." November 13, 2023, Future Trends in Polynomial Optimization, LAAS-CNRS, Toulouse, FR. [link]
4) "Frequency Domain Identification via Sum-of-Rational Optimization." September 25, 2023, European Research Network System Identification (ERNSI), Stockholm, SE. [link]
5) "Safety Analysis and Control using Measures." April 13, 2023, RISE 2023, Northeastern University. [link]
6) "Safety Analysis and Control using Measures." February 27, 2023, PhD Research Expo, Northeastern University. [link]
7) "Diameter Constrained Minimum Spanning Graphs." January 31, Current Themes of Discrete Optimization: Boot-camp for early-career researchers, ICERM. [link]
8) "Safety Analysis using Distance Estimation and Measures." August 24, 2022. CLEVR-AI MURI Yearly Review Meeting, Northeastern University. [link]
9) "Exploiting SDP Structure Yields Tighter Approximations." April 9, 2020. RISE, Northeastern University (remote). [link]
10) "Exploiting SDP Structure Yields Tighter Approximations." February 24, 2020. IPAM Control, Learning and Optimization workshop, University of California, Los Angeles. [link]
11) "Chordal Decompositions in Rank Minimized SDPs." May 30-31, 2019. Learning for Decision and Control (L4DC), Massachusetts Institute of Technology. [link]
12) "Chordal Decompositions in Rank Minimized SDPs." May 10, 2019. New England Machine Learning Day, Northeastern University. [link]
13) "Scattered data interpolation through B-spline wavelets and the Elastic Net." April 14, 2017. RISE, Northeastern University. [link]
14) "A parallelized Python-based Multi-Point Thomson Scattering analysis in NSTX-U." October 29, 2014. 56th Annual APS Plasma Physics Conference, New Orleans. [link]